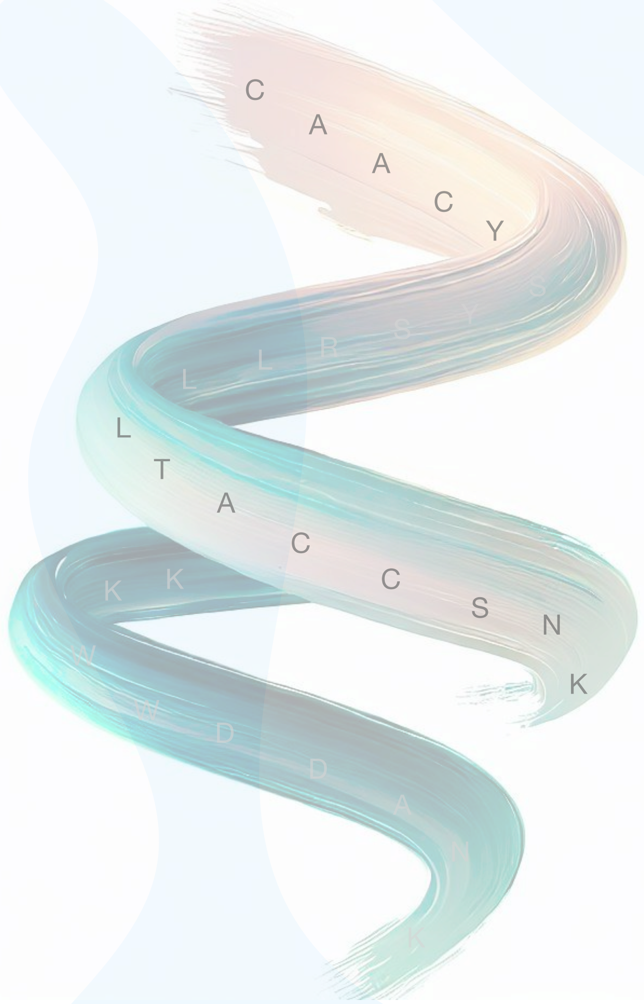


Protein Language Modelling

Deep Learning Summer School 2026

Alberto Santos and Ida Kristine Sandford Meitil —
Multi-omics Network Analytics Group

20.08.2026

Outline

- **Introduction**
- **Basic Concepts**
  - Understand the fundamental concepts of pLMs
  - Explore their architectures and how they are trained
- **Data Preparation**
  - Building Training and Test Datasets
  - Data Resources
- **Downstream Tasks**
  - Feature Extraction
  - Fine-tuning pre-trained models
- **Applications of pMLs**
- **Benchmarking**
- **Interpretability**
- **Challenges and Limitations**
- **Future directions**
- **Practical**

Introduction

Multi-omics Network Analytics

DTU Health Technology — Bioinformatics Section

Multimodal Data

Implementing tools to process, integrate, and analyse multimodal data. Diving into the benefits of harmonising multimodal data that converge to provide a comprehensive view of complex biological systems. Specifically we are interested in high-throughput multi-omics data generated using Mass spectrometry technology (proteomics and metabolomics) and metaomics data (metagenomics and metaproteomics).

Knowledge Graphs

Building High-quality Knowledge Graphs. Using and developing Knowledge Graph technologies and methods to structured data and to connect them to existing biological knowledge. These structures facilitate analysis and interpretation of complex data. We are contributing to a groundbreaking field by developing tools and methods to build, assess and investigate Knowledge Graphs and applying them to solve challenges in biology and health.

Graph Machine Learning

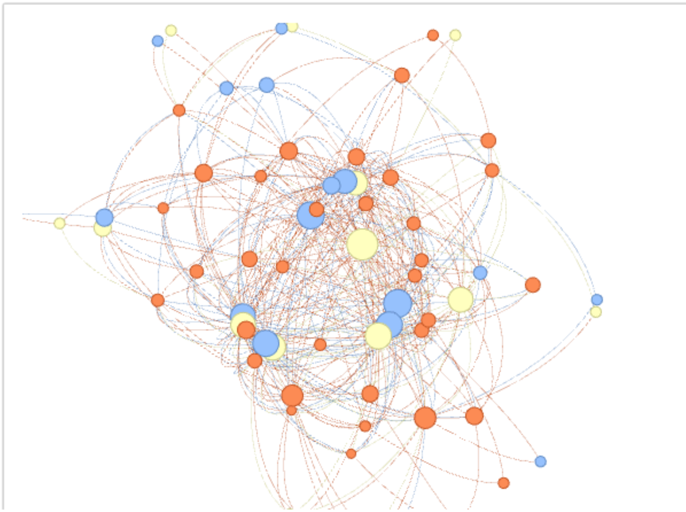
Developing and Applying Novel Methods on Graphs. Unleashing the power of Machine Learning on Graphs, a cutting-edge approach to extracting valuable insights from network data. We explore how this fusion of machine learning and graph theory helps to recognize patterns, generate predictions, and discovering new knowledge across a multitude of applications, including biological and medical networks.

Open Science

Data Science Democratisation. Focusing on data literacy training as a means to reduce inequality, and promoting open science by making all research, data content, and software open and accessible.

Microbial Communities

Exploring Microbial Communities and their Environments. Integrating multiple biological resources to unravel the assembly, interaction and adaptation mechanisms of microbial networks, offering insights into their functions and inpact on ecosystems, and how changes affect those communities.

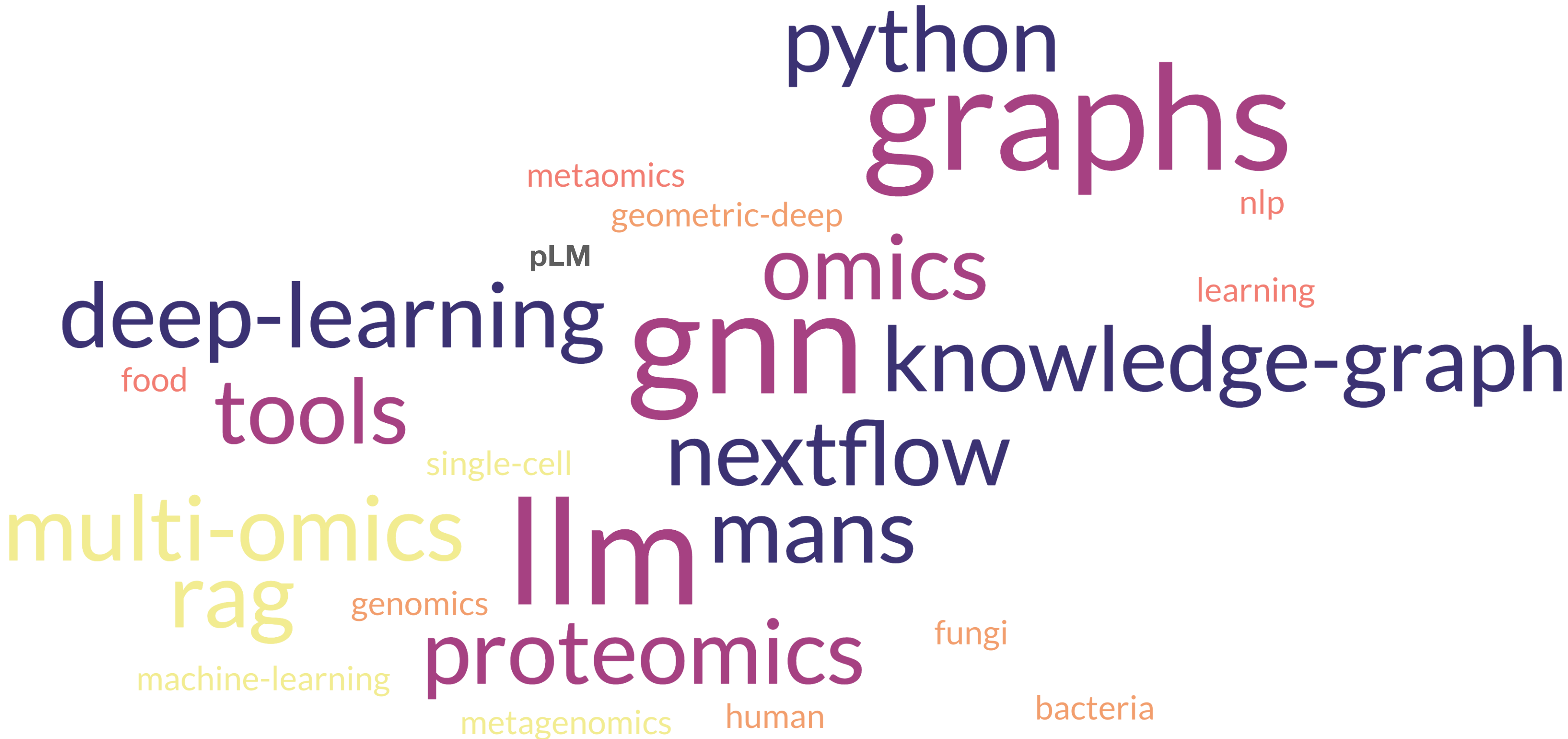


Clinical Computational Omics

Providing tools for the analysis and interpretation of clinical omics data. Integration of high-throughput omics data with computational and bioinformatics approaches to advance precision medicine and disease research. These projects aim to identify biomarkers, uncover disease mechanisms, and tailor treatments based on individual molecular profiles.

https://multiomics-analytics-group.github.io/

Research



python
graphs
metaomics
geometric-deep
nlp
pLM
omics
learning
deep-learning
gnn
knowledge-graph
food
tools
single-cell
nextflow
multi-omics
llm
mans
rag
genomics
fungi
proteomics
machine-learning
bacteria
metagenomics
human

Basic Concepts

Protein Language Models (pLMs)

- **pLMs** are deep learning architectures that learn the "grammar" and "semantics" of the **protein language**

- **Large Language Models (LLMs)** learn patterns from **text**, **pLMs** learn from large datasets of **protein sequences**

- **Represent complex biological information** within protein sequences in a machine-readable format (**embeddings**) and use them for **different biological downstream tasks**

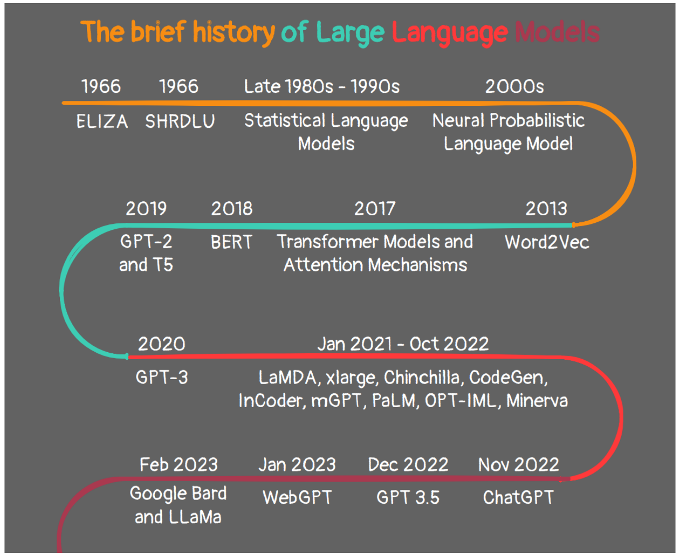
Billions of protein sequences known, only a tiny fraction have known functions or structures

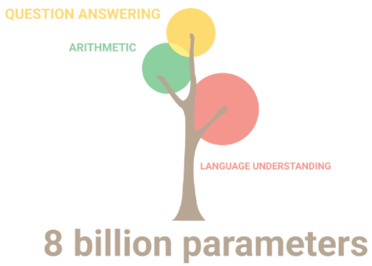
Experimental characterisation of proteins (structure, function, and interactions) is slow, expensive, and labor-intensive.

Traditional methods rely on homology, which struggles with distantly related or novel proteins

Language Models (LMs)

- **Language models (LMs)** enable analysing **patterns in language** by predicting words
- **Best performing models** are based on **Transformer architectures** and using **Attention mechanisms** (LLMs use a decoder-only variant of this architecture)
- LMs are trained using **self-supervised** learning:
    - **Masked self-attention**: looks at other tokens in the sequence (i..e, those that precede the current token).
    - **Feed-forward transformation**: transforms each token representation individually.
- Applications:



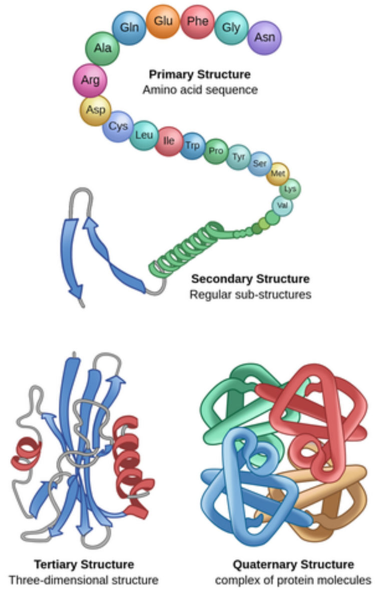


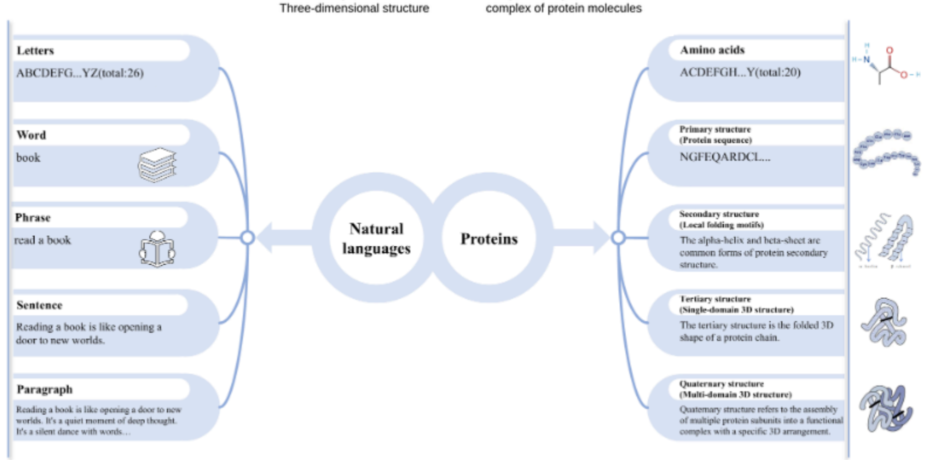
Attention Is All You Need
Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin
arXiv:1706.03762 [cs.CL] https://doi.org/10.48550/arXiv.1706.03762
https://techblog.ezra.com/an-overview-of-different-transformer-based-language-models-c9d3adafead8
https://tinyurl.com/LMtimeline
https://github.com/Hannibal046/Awesome-LLM/tree/main

The Language of Proteins

- Proteins play a **crucial role** in the **structure**, **function**, and **regulation of cells** and **tissues** within living organisms — Building blocks
- **Vocabulary** consists of **20 amino acids (AAs)** - letters

  The amino acid sequence **MGSA** would be tokenized into **[M]**, **[G]**, **[S]**, **[A]**
- These amino acids form **secondary structural elements** (e.g., motifs and domains)- words
- Full **sequence** — sentence
- **Structure, function and context** - meaning
- Differences with natural language:
  - No punctuation no differentiation of words, sentences and paragraphs
  - High variability in length
  - Text has fewer distant interactions — proteins 3D structure
  - Letters of proteins can be modified (e.g., post-translational modifications)





A Comprehensive Review of Protein Language Models
Lei Wang, Xudong Li, Han Zhang, Jinyi Wang, Dingkang Jiang, Zhidong Xue, and Yan Wang.
arXiv:2502.06881v1 [q-bio.BM] 08 Feb 2025

The language of proteins: NLP, machine learning & protein sequences
Dan Ofer, Nadav Brandes, Michal Linial.
Comput Struct Biotechnol J. 2021; 19: 1750–1758, 2021. https://doi.org/10.1016/j.csbj.2021.03.022

https://theory.labster.com/protein-structure/

Architectures

- A dominant architecture — **Transformers**
- Key component — **Self-Attention Mechanism:**
    - Amino acids that are far apart in the linear sequence can be very **close in the 3D folded structure** and functionally interact
    - Transformers use self-attention to model **long-range residue–residue interactions** (biological context).

Example: When encoding the amino acid Cysteine (C), self-attention can identify if another 'C' is nearby (important for disulfide bonds) or if specific hydrophobic residues are present far away but contributing to form a pocket

Transformer-based deep learning for predicting protein properties in the life sciences
Abel Chandra, Laura Tünnermann, Tommy Löfstedt, and Regina Gratz.
January 2023 eLife 12:e82819 DOI: 10.7554/eLife.82819

Pre-training Objectives

- **Self-Supervised Learning** PLMs are pre-trained on massive amounts of unlabelled protein sequence data.
- They learn by solving "fill-in-the-blank" or "predict-the-next-word" type tasks
- **Masked Language Modelling (MLM)**:

  A portion of amino acids in a sequence are randomly masked (replaced with a [MASK] token, a random amino acid, or left unchanged). The model's task is to predict the original, masked amino acids based on their surrounding context. ESM-2 (Lin et.al 2022) and ProtBERT (Brandes et.al 2022) are the examples of the models.

  Example:

  Sequence: ARSVL

  Masked: [A]-[R]-[MASK]-[V]-[L]

  Model predicts: **S**

  Forces the model to learn **bidirectional context**

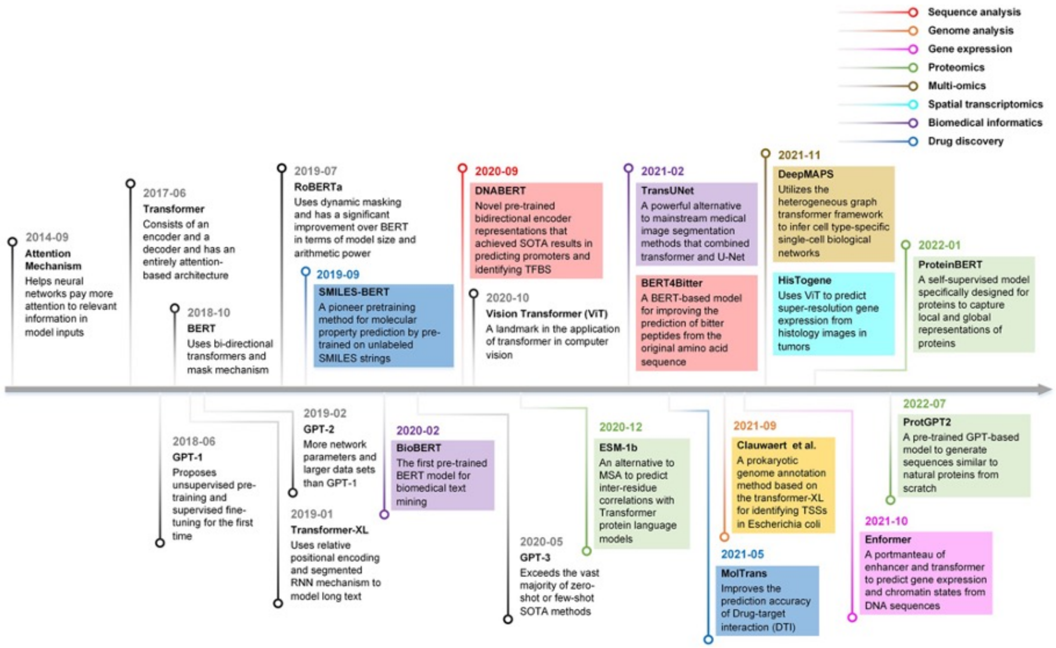
- **Next Token Prediction (NTP) / Autoregressive Modelling:**

  The model is trained to predict the next amino acid in a sequence, given all the preceding amino acids. ProGen (Madani, et.al 2020) , ProGen2 (Nijkamp et.al 2022), and ProtGPT2 (Ferruz, et.al 2022) follow this approach.

  Example:

  Given M-G-S-A-P, predict **K** as the next amino acid

  Learns **unidirectional context**, useful for **generating new sequences**



Applications of transformer-based language models in bioinformatics: a survey
Shuang Zhang, Rui Fan, Yuti Liu, Shuang Chen, Qiao Liu, Wanwen Zeng.
Bioinformatics Advances, Volume 3, Issue 1, 2023, vbad001, https://doi.org/10.1093/bioadv/vbad001

https://medium.com/@anrizal05/protein-language-models-from-amino-acid-tokens-to-sequence-embeddings-e488e89a330e

Data Preparation

Data Preparation

Building Training and Test Sets for PLMs

Danger

- PLMs are **trained on massive public protein sequence databases** like UniProt, UniRef, and MGnify.
- The most common source of bias is due to:

Data leakage occurs when information from the test set "leaks" into the training set mainly due to **sequence similarity** (identical samples or highly similar homologs)

- Consequences:
  - Inflated performance metrics
  - Poor generalisation to novel protein families
  - Misleading benchmarks between models
- Main splitting strategies:
  - **Random**: Simplest, but highly prone to data leakage in protein tasks due to homologous proteins appearing in both train and test sets. Discouraged.
  - **Sequence identity-based clustering**: The most widely accepted method
  - **Family/Fold-based splitting**: Even stricter, ensuring generalisation to entirely new protein classes
- Always use a separate **validation set** for hyperparameter tuning, model selection, and early stop during training. This prevents overfitting to the test set and provides an unbiased estimate of model performance during development.

UniLanguage Datasets

- This is the first **protein dataset** with a **focus on language modelling**
- Analysis of UniProt database to define different properties that can bias or hinder the performance of pLMs such as homology, domain of origin, quality of the data, and completeness of the sequence.

Dataset and results

We provide a script to obtain train, validation and test sets for all domains, qualities and protein completion.

Script usage

```
python get_data.py --domain [domain] --complete [complete] --quality [quality]
```

Parameters:

- Domain: Eukarya= `euk` , Bacteria= `bac` , Archaea= `arc` , Virus= `vir`
- Complete: Complete proteins= `full` , Fragmented proteins= `frag`
- Quality: Experimental evidence= `exp` , Predicted= `pred`

Language modelling for biological sequences – curated datasets and baselines
Jose Juan Almagro Armenteros, Alexander Rosenberg Johansen, Ole Winther, Henrik Nielsen
doi: https://doi.org/10.1101/2020.03.09.983585

Data Resources

High-quality, comprehensive and freely accessible resource of **protein sequence and functional information**

RCSB Protein Data Bank (RCSB PDB) provides access and tools for exploration, visualisation, and analysis of:
Experimentally-determined 3D structures
Computed Structure Models (CSM) from AlphaFold DB and ModelArchive

Functional analysis of proteins by **classifying them into families and predicting domains and important sites.**

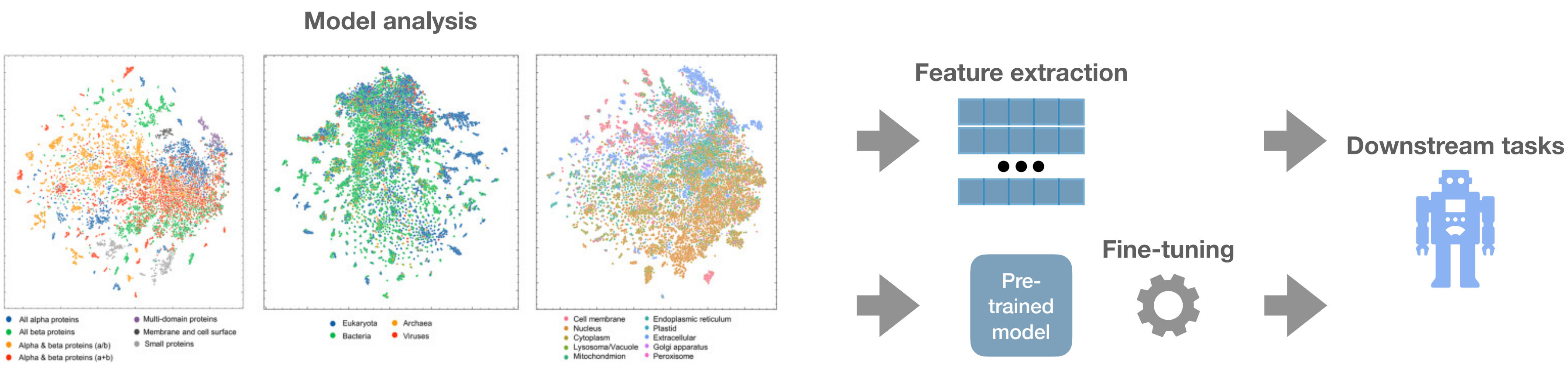
A database of **orthology** relationships and **functional annotation** based on **5090 organisms** and **2502 viruses**

IntAct provides a free, open source database and analysis tools for **molecular interaction data.**
It also includes **negative datasets**

Downstream Tasks

Post-training — Downstream Tasks

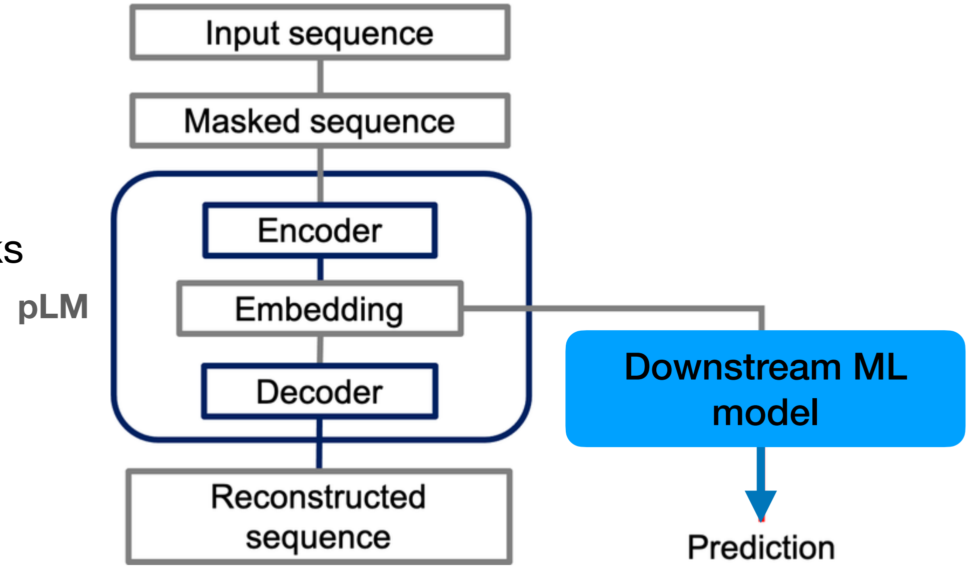
- After a pLM has been pre-trained on unlabelled datasets, models can be applied to various downstream tasks through different approaches:
  - **Model analysis** Analysing attention weights to identify functionally important regions, visualising vector representations or attention matrices to discover relationships between proteins
  - **Feature extraction** Using embeddings or attention matrices as inputs to train separate supervised models for tasks like expression profiles or protein-protein interactions
  - **Fine-tuning** Adapting the pre-trained model to specific tasks with labeled data, training on small datasets of protein variants with measured properties



Feature Extraction

Protein Embeddings: Representing Amino Acids Numerically

- Embeddings are dense, low-dimensional vector representations of amino acids or entire protein sequences
  - **Per-residue embeddings**: These are vectors for each individual amino acid in a protein sequence, capturing its context within that specific protein
  - **Per-protein (pooled) embeddings**: For downstream tasks requiring a single vector per protein, the per-residue embeddings are "pooled" (e.g., averaged or max-pooled) across the sequence length to create a single fixed-size vector representing the entire protein
- These numerical representations **capture semantic relationships** (e.g., biochemical properties, evolutionary relatedness) between amino acids and their context within the protein
- These embeddings are **learned during the pre-training phase**



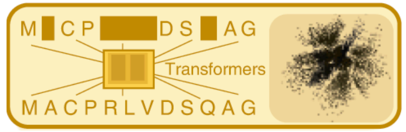
https://www.uniprot.org/help/embeddings

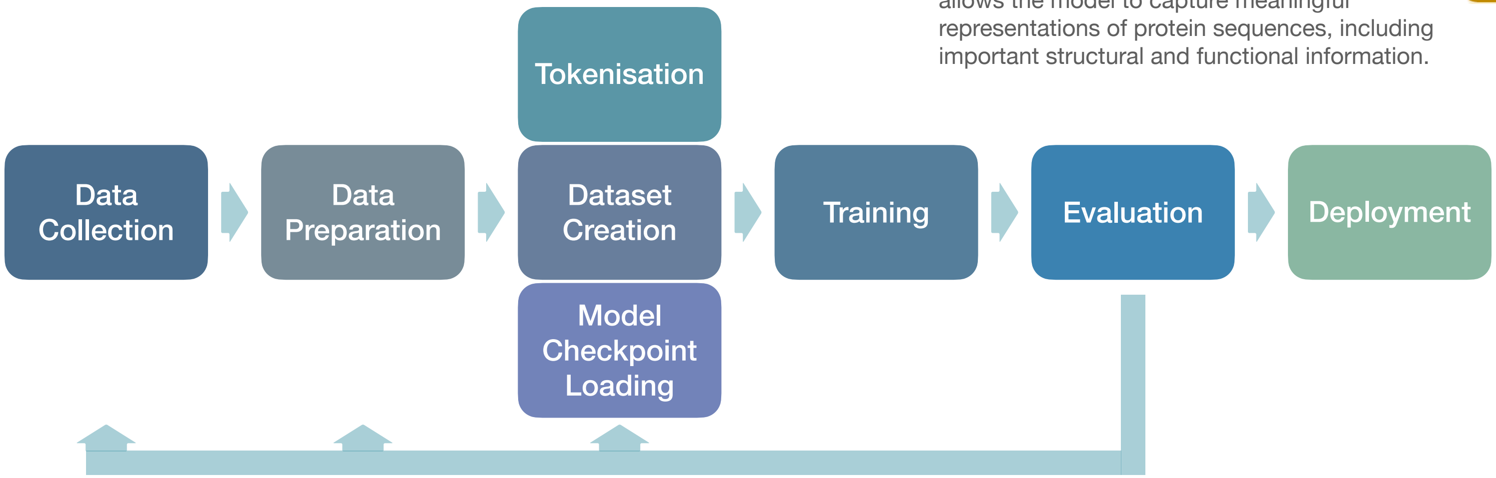
Fine-tuning

Fine-tuning pLMs

Fine-tuning refers to a learning technique where a **pre-trained language model** is further trained on a specific downstream task using task-specific data.
The goal is to adapt the knowledge captured by the pre-trained model to the specific characteristics of the target task, thereby improving the model's performance on that task.

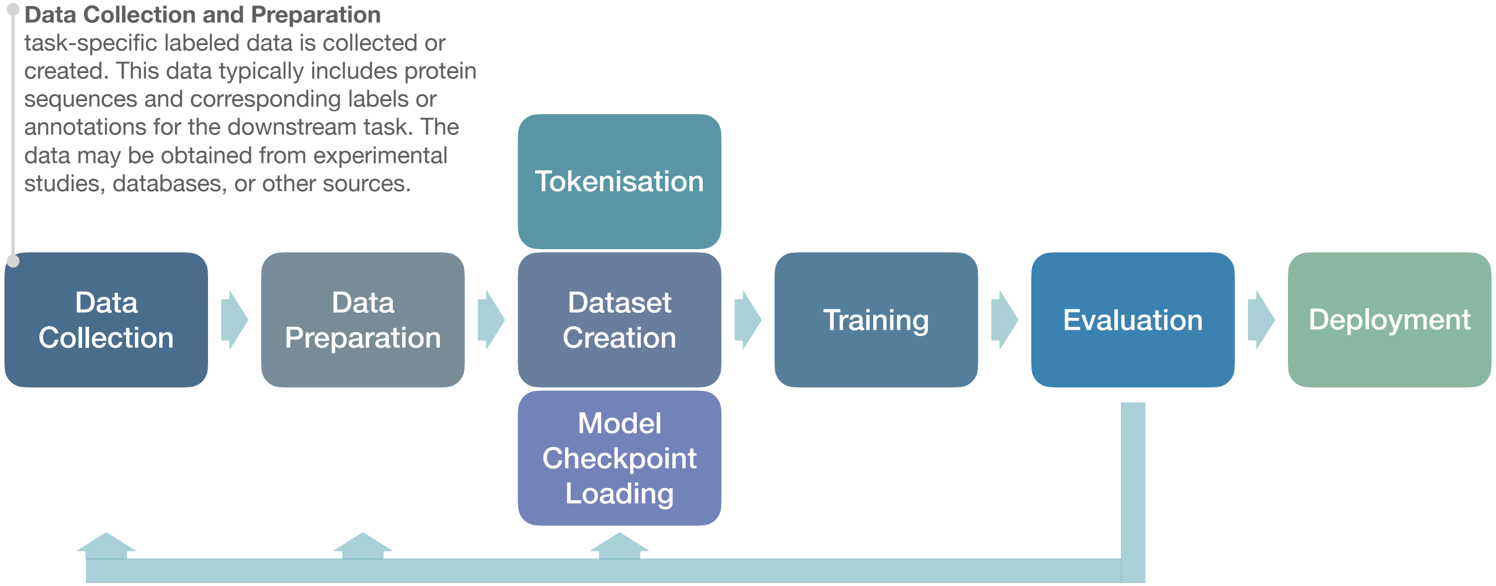
Pre-trained pLM is a model pre-trained on massive protein sequence databases. During pre-training, the models learn to predict missing parts of protein sequences based on the context provided by the surrounding sequences. This allows the model to capture meaningful representations of protein sequences, including important structural and functional information.





Single-sequence protein structure prediction using a language model and deep learning
Ratul Chowdhury, Nazim Bouatta, Surojit Biswas, Christina Floristean, Anant Kharkar, Koushik Roy, Charlotte Rochereau, Gustaf Ahdritz, Joanna Zhang, George M. Church, Peter K. Sorger & Mohammed AlQuraishi.
Nature Biotechnology (2022)

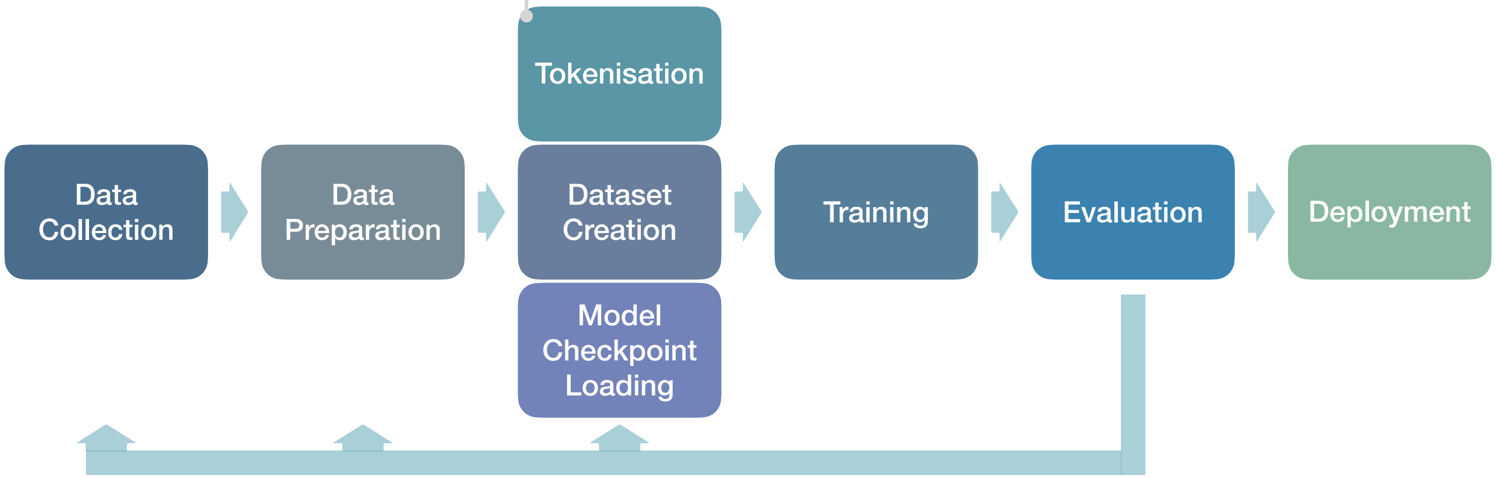
Fine-tuning pLMs



Fine-tuning pLMs

Tokenisation is a required task in NLP that divides text into smaller units called tokens. In NL, tokens can be either words, characters, or subwords. pLMs typically tokenise at character level (i.e., residue), so no need to train a custom tokeniser.

String:	The cat sat on the mat	MSTIYSTGKVCNP...
Possible tokenizations:	[*start*] [T] [h] [e] [*space*] [c] [a] [t] ...	[*start*] [M] [S] [T] [I] [Y] [S] [T] [G] ...
	[*start*] [The] [cat] [sat] [on] [the] [mat]	[*start*] [MS] [TI] [YS] [TG] ...
	[*start* The] [cat sat on] [the] [mat]	[*start* M] [STI] [YST] [GK] [VCN] ...



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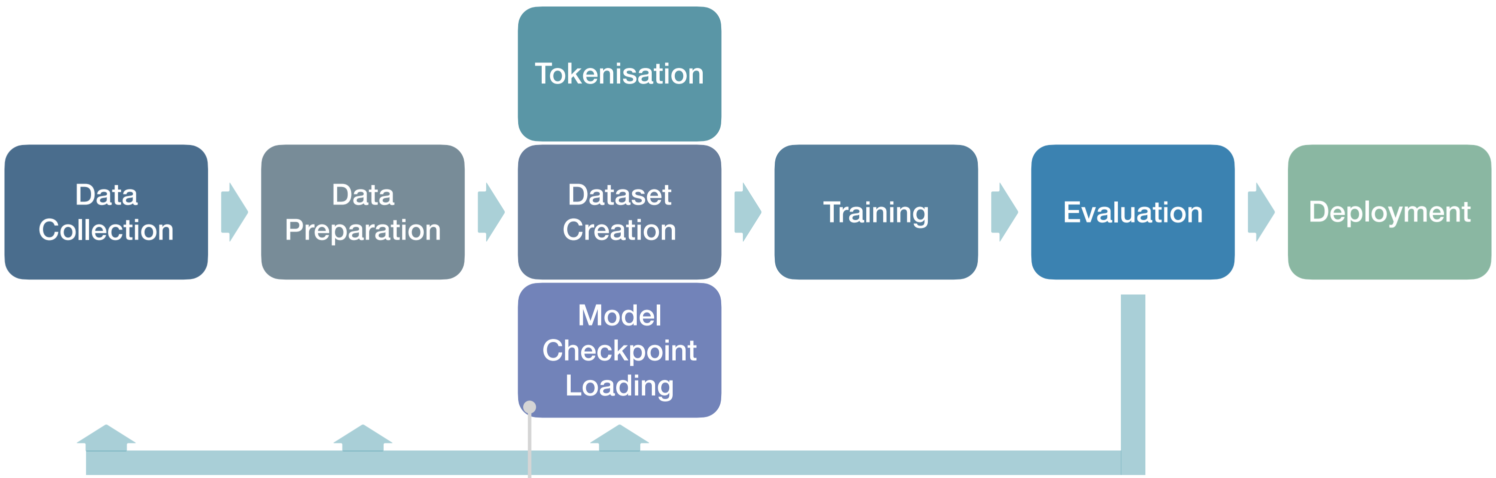
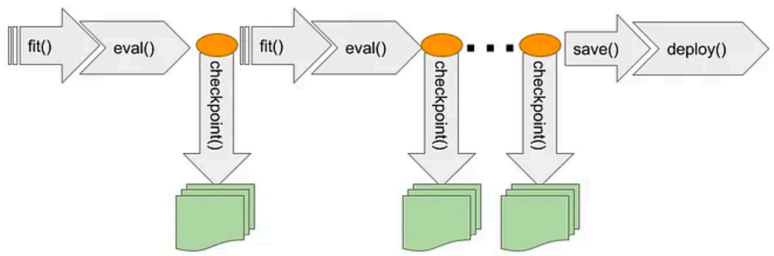
Dan Ofer, Nadav Brandes, Michal Linial.
Comput Struct Biotechnol J. 2021; 19: 1750–1758, 2021. https://doi.org/10.1016/j.csbj.2021.03.022

Fine-tuning pLMs



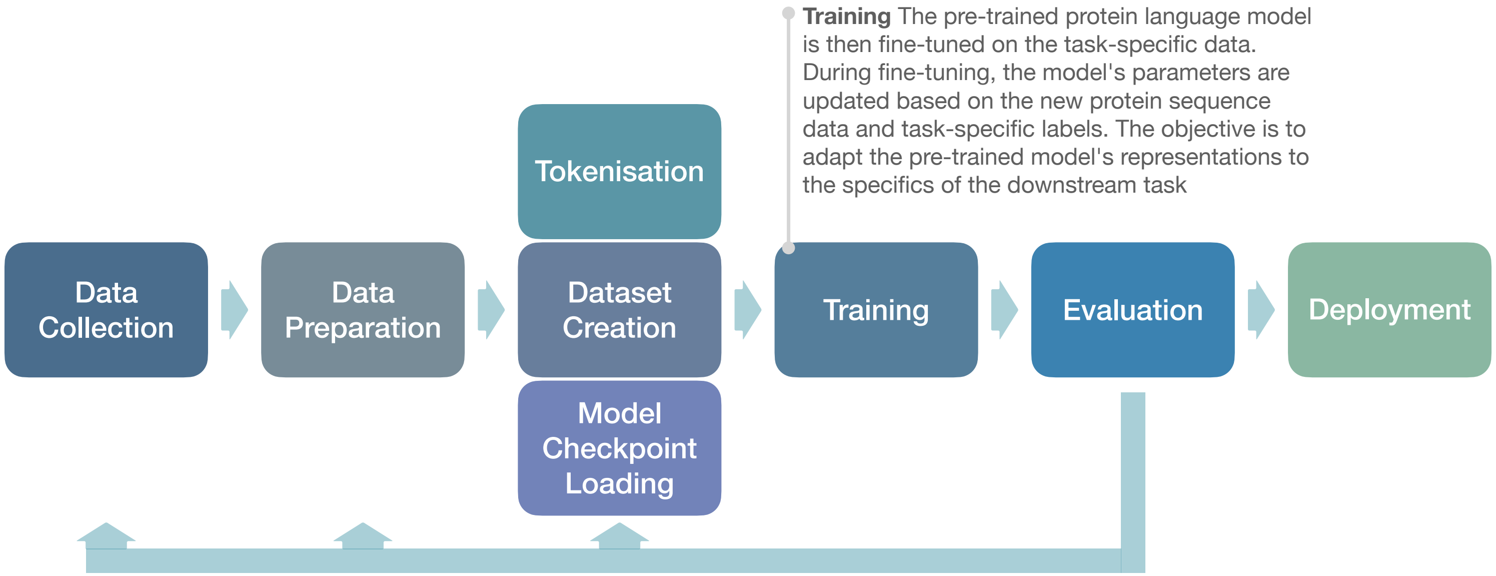
Checkpoint is an intermediate dump of a model's internal state — weights, current learning rate, parameters, etc. This allows to resume the training from that point. Many existing pLMs provide checkpoints that are ready to initialise (warm start) your own models and in many cases get better performance than a baseline model trained from scratch.



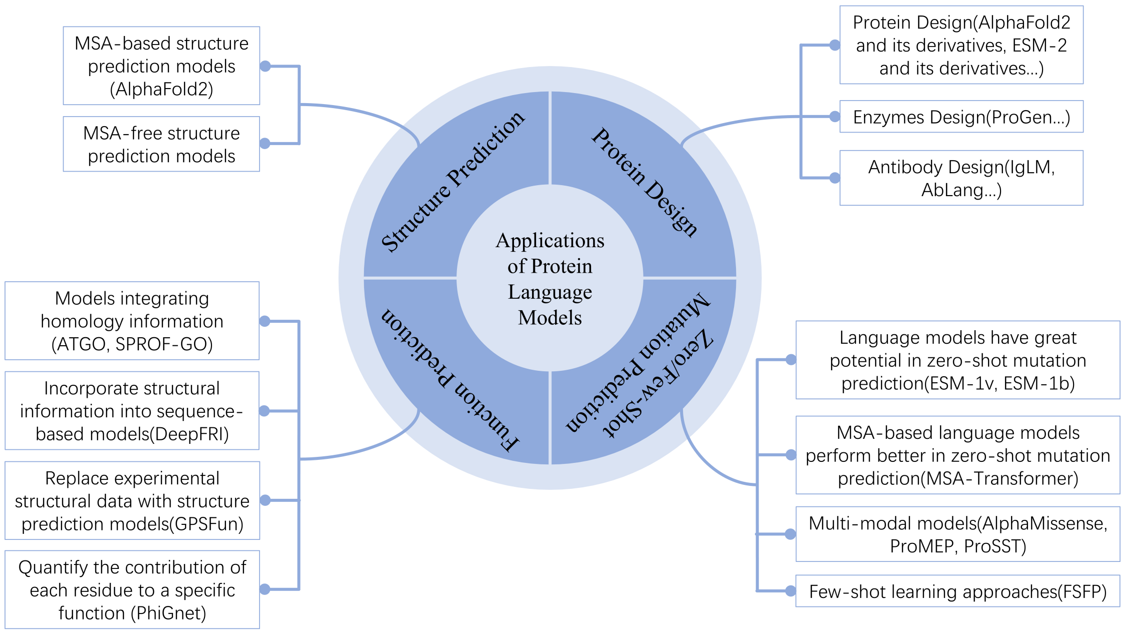
Exploiting pretrained biochemical language models for targeted drug design.
Gökçe Uludoğan, Elif Ozkirimli, Kutlu O Ulgen, Nilgün Karalı, Arzucan Özgür
Bioinformatics, September 2022, https://doi.org/10.1093/bioinformatics/btac482

https://towardsdatascience.com/ml-design-pattern-2-checkpoints-e6ca25a4c5fe

Fine-tuning pLMs



Applications of pLMs



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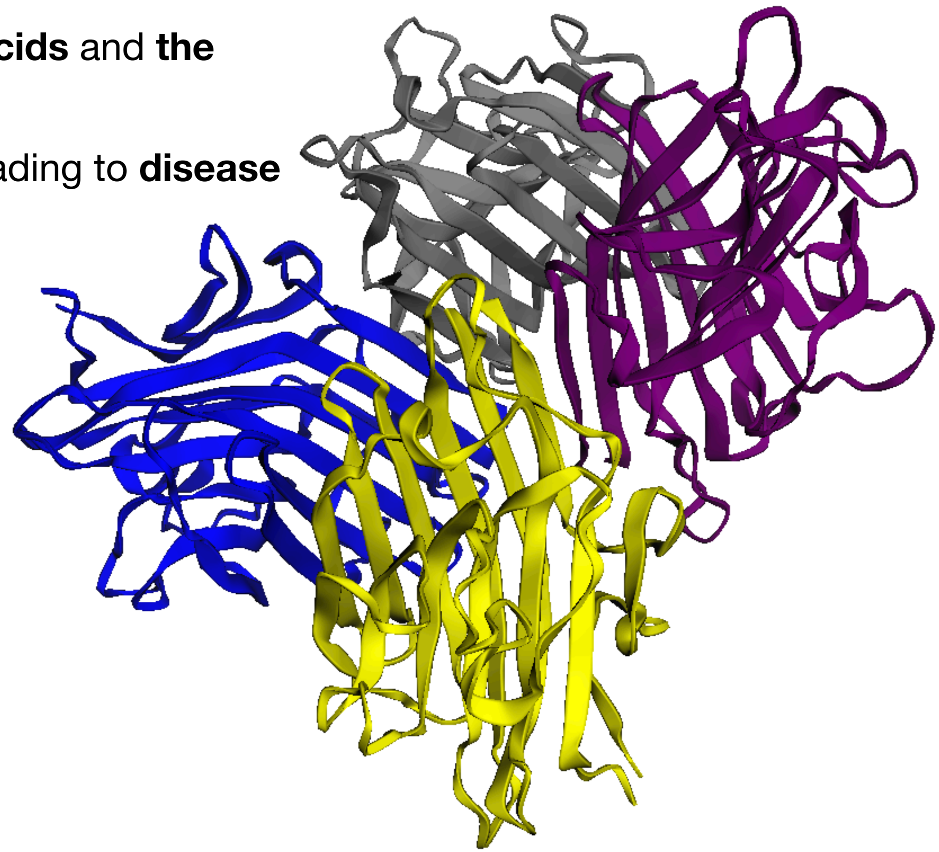
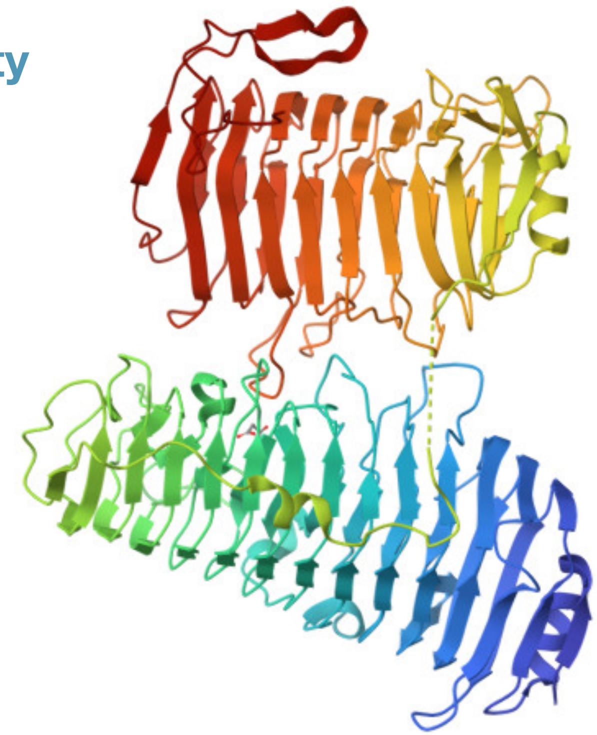
Lei Wang, Xudong Li, Han Zhang, Jinyi Wang, Dingkang Jiang, Zhidong Xue, and Yan Wang.
arXiv:2502.06881v1 [q-bio.BM] 08 Feb 2025

Prediction of **Protein Structure**

- The **function of a protein** depends on its **3D structure**
- The structure is determined by the **specific sequence of amino acids** and **the interactions between them**
- Any **changes** in the sequence can alter the shape and function, leading to **disease**
- **Remarkable success:**
  - Evolution-based: ESM
  - Single-sequence-based: RGN2
- Applications:
  - Function prediction
  - Drug design
  - Mutation impact

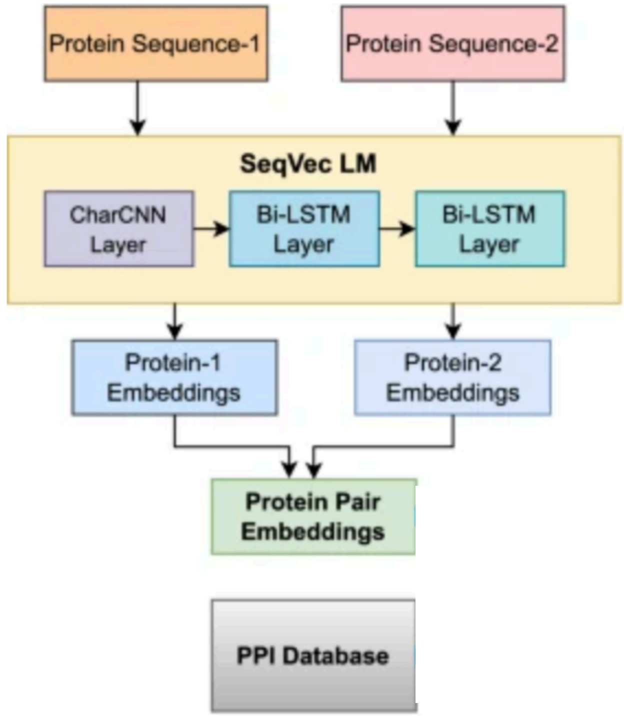


Protein Property Prediction



Prediction of **Protein-protein Interactions**

- Protein–protein interactions (PPI) is a **critical problem in biology**
    - Some **protein functions** in a cell depend on having **physical contact** with other proteins present
    - PPIs identified through **experiments cover only a small percentage** of PPI networks
    - The entire PPI network cannot be explored experimentally due to **costs and time requirements**



Graph-BERT and language model-based framework for protein–protein interaction identification
Kanchan Jha, Sourav Karmakar & Sriparna Saha. Scientific Reports volume 13, Article number: 5663 (2023)

Protein Engineering and Design

- Using PLMs to generate novel protein sequences with desired properties or to optimize existing proteins for improved function. This moves from "prediction" to "creation."
- Pre-trained transformer models enabled **text generation** including texts **with specific properties** such as style
- Fine-tuning pre-trained models on **protein families** will similarly enable the generation of **novel potentially functional sequences**.
- Examples:
  - **Additional information** such as cellular compartment or function can aid the **control over specific properties**
  - **De novo design: Generating entirely new enzymes with specific catalytic activities.**
  - **Therapeutic design: Engineering antibodies with enhanced binding affinity or stability for drug development.**
  - **Bioremediation: Designing enzymes that can degrade plastics or pollutants.**

Text generation	When I was living in Lisbon I was so shocked at the lack of diversity that everyone chose to go back home. No matter there background, it was always the same things. I was shocked at the amount of food options… it was like one-dish-and-you're-on-your-face food with almost no meat options.	Lisbon is a world-renowned destination for great culture, world-class dining, and an unparalleled opportunity for relaxation. It is also one of the safest cities in Europe, with an official crime rate that is half that of Paris or New York, …
Protein sequence generation	GCYVQAFGMAGCKINALNQYVINSLNYVTNVP YNLVGGKGYIYVWVLCGGNGGNGGNGAD AIPGGADGGGGAGDPGAVLMSFYAYYPGATVT IKVGNAGAAGGNGTNGGSGGTSTTVVNG TTYTFTGGTSGKSGGTGVSGSSGLPGVDSSGT TPGGGNGANGITTVFWS	MTQDDRKRYRQLSMGQELAQMCAECGARQS LTVGTSSQRFCSEQCAYAARNRSHKAATSR SQGLDPSRRQYLAVKRENNQFCVLCRAEGP PSHYDRPLAVDDHDHDHESGPPRRLLCNRC NLGPLGKFDDPALLRDAVAHLEGMAGRA

ProtGPT2 is a deep unsupervised language model for protein design
Noelia Ferruz, Steffen Schmidt & Birte Höcker. Nature Communications volume 13, Article number: 4348 (2022)

Benchmarking

Benchmarking

- Benchmarking evaluating the performance of different models on standardised datasets and tasks using agreed-upon metrics

Evaluation metrics

- **Classification**: Accuracy, precision, recall, F1-score, Area Under the Receiver Operating Characteristic curve (AUROC), Area Under the Precision-Recall curve (AUPRC) — e.g., protein function prediction

- **Regression**: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Spearman's correlation coefficient — e.g., stability prediction

- **Structure prediction**: Root Mean Square Deviation (RMSD), GDT_TS (Global Distance Test – Total Score), pLDDT (predicted Local Distance Difference Test – an AlphaFold metric).

Benchmarking Platforms

- **Tasks Assessing Protein Embeddings (TAPE)**: benchmark of five diverse protein tasks — remote homology, contact prediction, secondary structure, fluorescence, stability (https://github.com/songlab-cal/tape)
- **ProteinGym**: Collection of benchmarks for evaluating how well models predict the functional impact of protein variants (https://proteingym.org/)
- **Functional Landscape of Interacting Proteins(FLIP)**: A benchmark specifically designed for evaluating protein fitness prediction models in scenarios with limited data, focusing on constrained and challenging settings (https://www.biorxiv.org/content/10.1101/2021.11.09.467890v2)
- **Protein Representation Benchmark (PROBE)**: A benchmarking framework for assessing protein representations on various function-related prediction tasks (semantic similarity, ontology-based function prediction, and drug target classification) (https://codeocean.com/capsule/8584011/tree/v2)

Interpretability

Interpretability

Opening the Black Box

Polysemanticity

Protein Language Models (PLMs) map hundreds of distinct biological concepts onto a small number of neurons.

This creates "**polysemantic**" neurons (superposition), where a single neuron activates for multiple, entirely unrelated structural or functional traits.

The Monosemantic goal

Mechanistic interpretability seeks to reverse-engineer these models. The goal is to **isolate individual concepts**.

Aims to decompose these complex representations into **clear, single-concept** (**monosemantic**) features, mapping directly to specific biological logic.

Sparse Autoencoders (SAE)

Decomposing Dense Representations

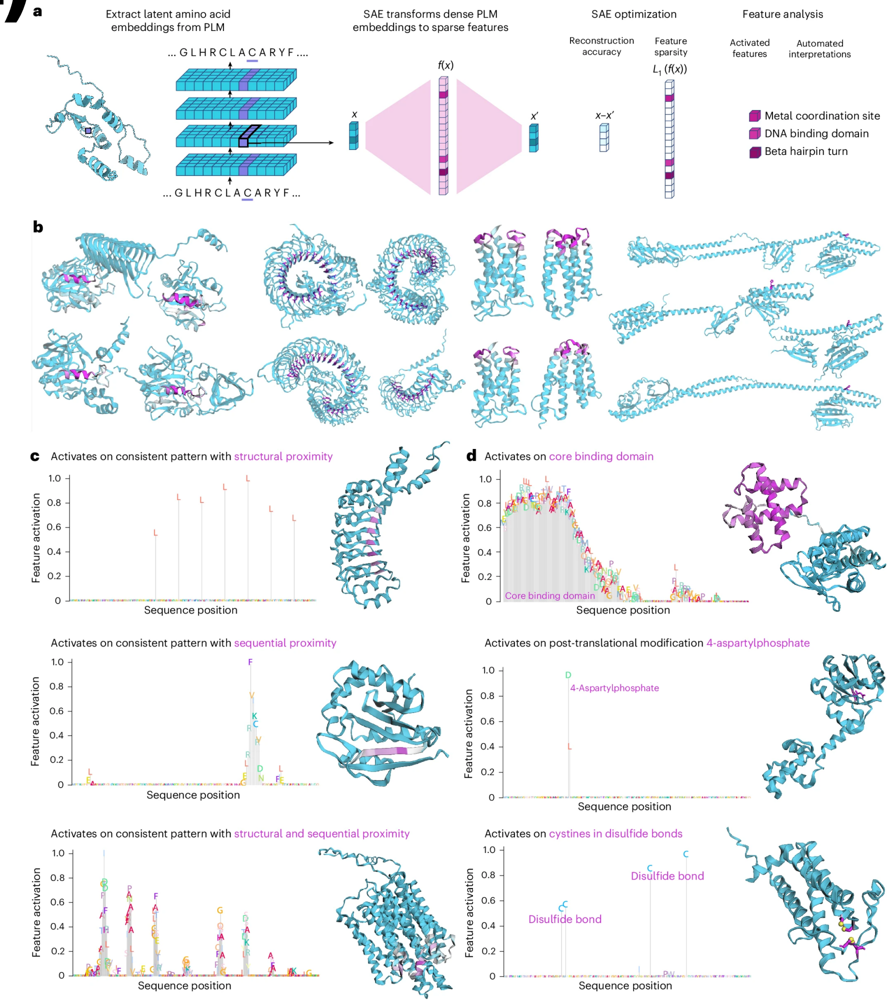
- An unsupervised neural network designed to learn a dictionary of fundamental features from unlabelled data.

- Resolve superposition by expanding dense layer activations into a much higher-dimensional, sparse latent space.

- The Sparsity Constraint: Unlike traditional autoencoders that rely on a low-dimensional bottleneck, SAEs enforce an information bottleneck through activation sparsity. Most neurons in the hidden layer must remain inactive (zero) for any given input.

- Obejective:
  - **Disentanglement**: Forces the network to learn statistically independent, robust features rather than entangled, dense vectors.
  - **Interpretability**: SAEs are used to decompose opaque embeddings into linear combinations of human-interpretable, "monosemantic" concepts.

InterPLM (Simon & Zou, 2024) showed that SAE applied to ESM2's internal activations discovers latents that correspond to biologically meaningful concepts such as binding sites, structural motifs, and specific amino acid types.



https://www.nature.com/articles/s41592-025-02836-7

SAE Architecture

- **Encoder**: Maps the input to a high-dimensional, sparse latent representation:

$$h = \mathrm{ReLU}(W_{enc}x + b_{enc})$$

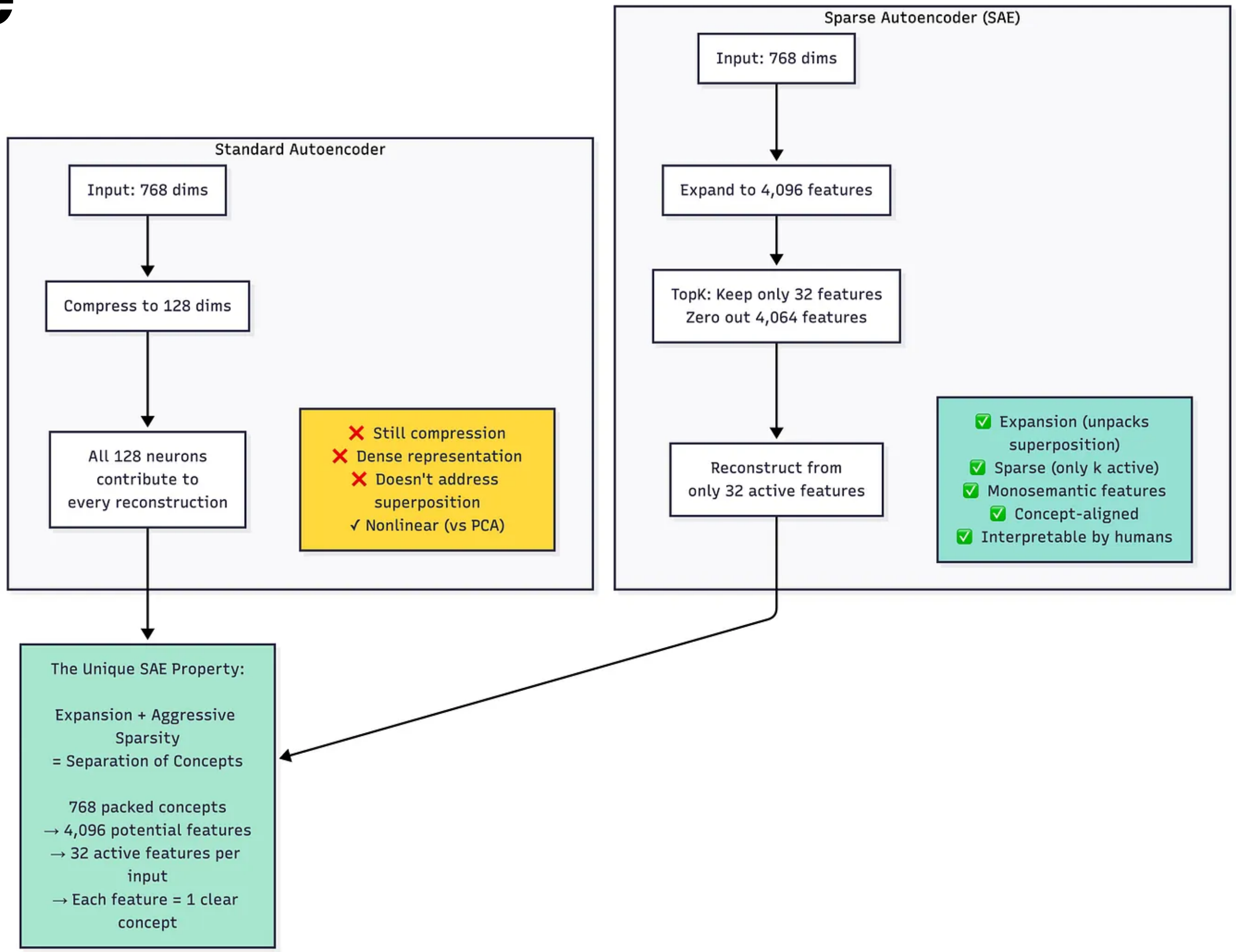
- **Decoder**: Reconstructs the input from the sparse latent vector to produce *x*'.

$$x' = W_{dec}h + b_{dec}$$

- **The Loss Function**:
  - The loss is a combination of the standard reconstruction error and a sparsity penalty.

$$L = \|x - x'\|_2^2 + \lambda \|h\|_1$$

  - **Reconstruction Loss** $\|x - x'\|_2^2$: Ensures the learned features retain the necessary information to rebuild the original input (typically Mean Squared Error).
  - **Sparsity Penalty** $\lambda\|h\|_1$: L1 regularisation applied directly to the activations forcing the majority of *h* to be zero.



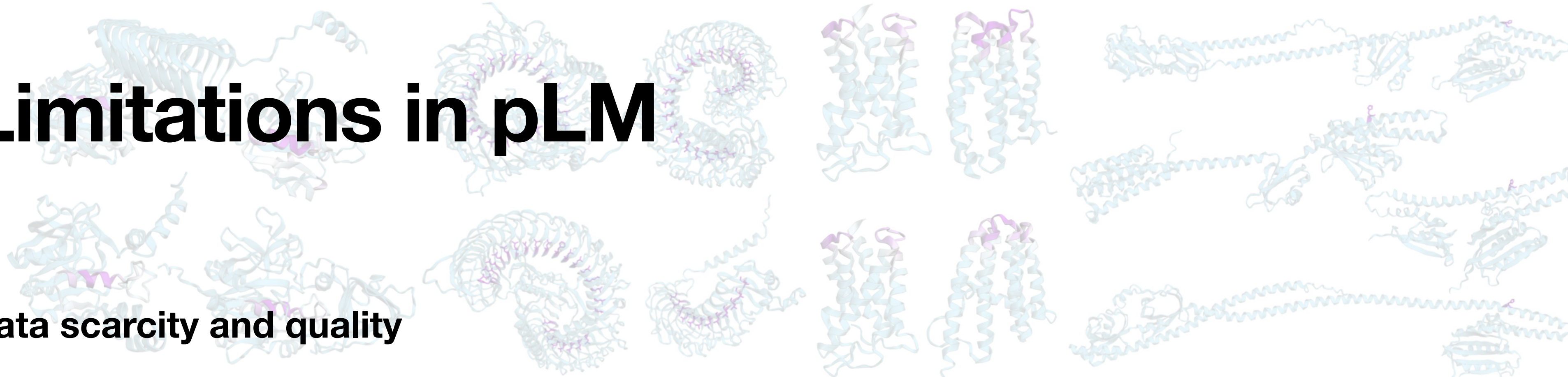
https://medium.com/data-science-collective/building-sparse-autoencoders-saes-from-scratch-b93c1a72e0ac

SAE’s Important Parameters

- **L1 coefficient (λ)**:
  - Controls the trade-off between reconstruction fidelity and sparsity.
  - High λ: Extreme sparsity, highly interpretable but poor reconstruction (high MSE).
  - Low λ: Excellent reconstruction, but dense features.
- **Expansion factor / Dictionary size (*n*)**:
  - The ratio of the hidden layer dimension to the input dimension.
  - Larger dictionaries capture more granular, monosemantic features but require significantly more data to train without dead neurons.
- **Neuron resampling / Initialization**:
  - Techniques required to re-initialize "dead neurons" (neurons that never activate across the dataset due to the L1 penalty) during training to maximize the utility of the dictionary.

Limitations

Limitations in pLM



Data scarcity and quality

- **Lack of high-quality, experimentally validated data** such as structures, binding affinities, and functional assays. This limits the ability to fine-tune PLMs for specific, accurate predictions.
- **Protein databases are biased**; some protein families or organisms are over-represented. This can affect generalization to under-represented regions of protein space.

Interpretability

- **Prediction results are still difficult to interpret**.

Computational resources

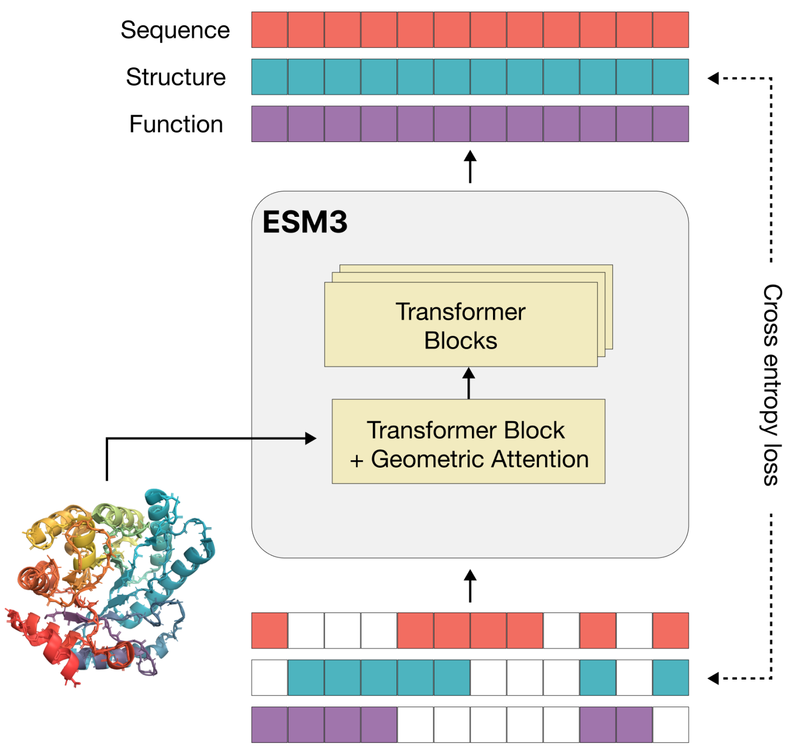
- **Training SOTA models requires hundreds to thousands of GPUs**, making them inaccessible to many researchers and labs.

Future Directions

- **Multi-modal pLMs** Integrating different types of biological data beyond just sequence (e.g., 3D structural information, experimental function data, interaction networks)

- **Improved architectures** Smaller, more efficient models with comparable performance, reducing computational costs and increasing accessibility.

- **Better pre-training strategies** More sophisticated self-supervised learning objectives that explicitly encode deeper biological priors or physical constraints into the models.



Simulating 500 million years of evolution with a language model
Thomas Hayes, Roshan Rao, Halil Akin, Nicholas J. Sofroniew, Deniz Oktay, Zeming Lin, Robert Verkuil, Vincent Q. Tran, Jonathan Deaton, Marius Wiggert, Rohil Badkundri, Irhum Shafkat, Jun Gong, Alexander Derry, Raul S. Molina, Neil Thomas, Yousuf A. Khan https://orcid.org/0000-0003-0201-2796, Chetan Mishra, Carolyn Kim, Liam J. Bartie, Matthew Nemeth, Patrick D. Hsu, Tom Sercu, Salvatore Candido, and Alexander Rives
DOI: 10.1126/science.ads0018

Recent Advances

- **Multimodal grounding**: Early PLMs (e.g., ESM-2, ProtT5) relied on primary amino acid sequences. Modern models embed structural topology (3Di states or coordinate vectors) and function directly into tokenisation, eliminating the need to infer 3D geometry solely from sequence context.
- **Modernized compute primitives**: Replacing GELU, absolute positional encodings, and standard LayerNorm with SwiGLU, RoPE, and RMSNorm yields severe efficiency gains and stability.
- **Unified Objectives**: Earlier workflows required separate models for feature extraction (e.g., ESM-2) versus sequence generation (e.g., ProGPT). Unified frameworks handle zero-shot property prediction and generative design within a single checkpoint.

Feature / Task	ESM2	ESM-c	ESM3
Primary Role	Sequence embeddings & property prediction	State-of-the-art sequence representations	Multimodal generation & de novo design
Architecture	Legacy BERT-style Transformer	Modernized Transformer (RoPE, SwiGLU, RMSNorm)	Multi-track generative Transformer
Input Modalities	Primary sequence	Primary sequence	Sequence, 3D coordinates, function tokens
Compute Efficiency	Baseline baseline	Efficiency gain over ESM2	High requirements
Licensing / Access	Open weights (Apache 2.0)	Open weights (300M/ 600M MIT license)	Commercial API / Restricted weights

Practical

https://tinyurl.com/2u4v8sme

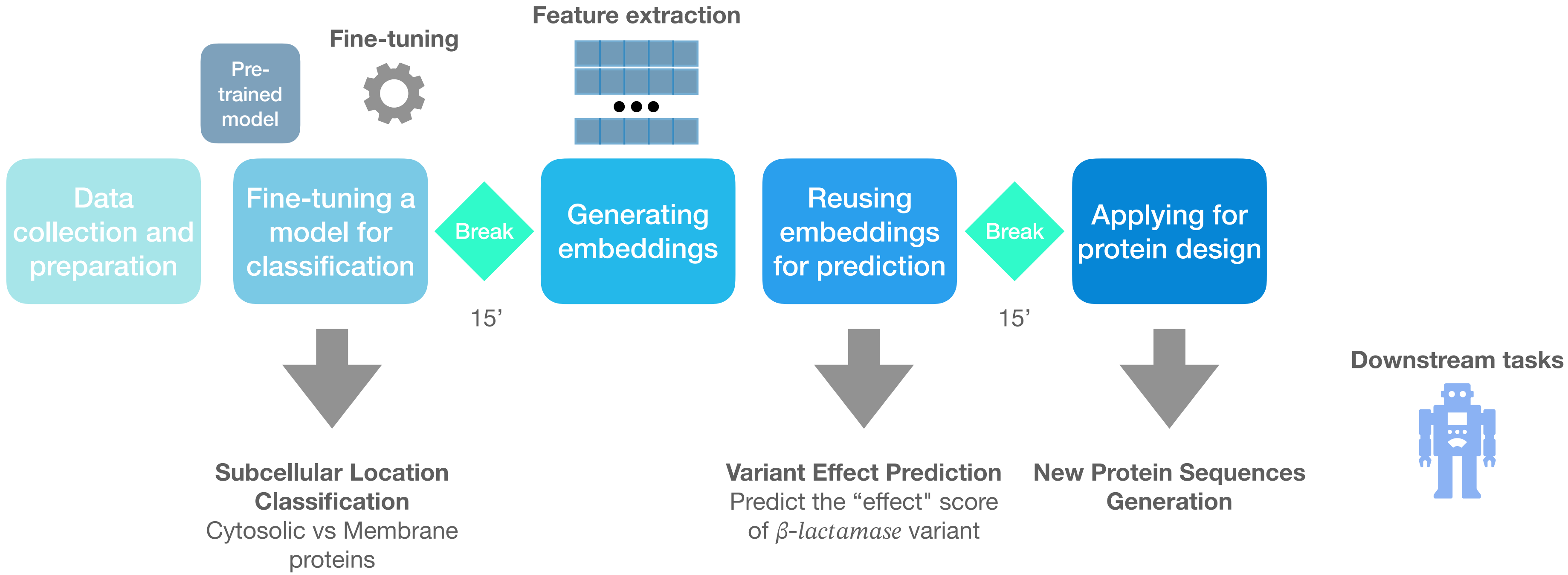
Course Structure

Theory		Link	Data	Models
	Topics	slides		
Hands-on				
	Sequence Analysis	Seq. Analysis Notebook Open in Colab	Exploring protein sequence and structure data	
	Fine-tuning a model	Model Training Notebook Open in Colab	Taking an existing model and tuning it for other prediction/classification tasks	Evolutionary Scale Modeling (ESM)
	Working with Embeddings	Embeddings Notebook Open in Colab	Accessing Protein representations (embeddings) generated by existing LMs	ProTrans
	Predictions	pML Predictions Notebook Open in Colab	Using embeddings for predicting features or classifying sequences	ProTrans
	Protein Design	Protein Design Notebook Open in Colab	*De novo* protein design and engineering using a LLM	ProtGPT2, ESM
	Mechanistic Interpretability	Interpretability Notebook Open in Colab	Opening the black box: training a Sparse Autoencoder on ESM2's internal activations to find human-interpretable features and checking whether they track known β-lactamase biology	ESM2

SCAN ME

Practical

https://tinyurl.com/2u4v8sme



Thank you

DTU

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